

NEW CARLISLE, QC, Canada

WMO: 716190

Lat: 48.0119N

Lon: 65.3317W

Elev: 152

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.4	(c) -3.3	(d) -19.9	(e) 1.9	(f) -6.3	(g) -14.5	(h) 2.5	(i) -2.3	(j) 29.9	(k) 9.3	(l) 27.4	(m) 10.1	(n) 12.3	(o) 290	(p) 0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.9	(c) 79.1	(d) 67.8	(e) 76.0	(f) 65.7	(g) 73.2	(h) 64.3	(i) 70.0	(j) 76.3	(k) 68.0	(l) 73.5	(m) 66.2	(n) 71.0	(o) 7.3	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.6	101.9	73.4	65.9	96.0	70.9	64.3	90.6	69.2	34.1	76.4	32.4	73.6	31.1	71.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.5	(b) 20.5	(c) 17.9	(e) -12.6	(f) 85.7	(g) 4.4	(h) 3.3	(i) -15.8	(j) 88.1	(k) -18.4	(l) 90.0	(m) -20.9	(n) 91.8	(o) -24.1	(p) 94.2		
(a) 23.5	(b) 20.5	(c) 17.9	(e) -13.3	(f) 74.1	(g) 4.3	(h) 1.8	(i) -16.5	(j) 75.3	(k) -19.0	(l) 76.4	(m) -21.4	(n) 77.3	(o) -24.6	(p) 78.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.1	14.8	16.0	24.7	35.8	46.8	57.0	64.3	63.2	55.6	44.8	34.0	22.7
	DBStd	18.68	10.30	9.91	8.87	6.18	6.29	6.40	4.88	4.81	6.20	6.32	7.55	8.78
	HDD50	4942	1093	952	785	429	141	11	0	0	17	186	481	847
	HDD65	9191	1558	1372	1250	877	563	251	71	91	289	627	930	1312
	CDD50	1332	0	0	0	2	43	222	444	411	184	25	1	0
	CDD65	105	0	0	0	0	0	12	50	37	6	0	0	0
	CDH74	508	0	0	0	0	8	84	229	153	34	0	0	0
	CDH80	71	0	0	0	0	0	11	33	23	4	0	0	0
Wind	WSAvg	7.7	9.9	9.3	8.6	7.9	6.8	6.2	5.9	6.0	6.6	7.8	8.5	9.4
Precipitation	PrecAvg	40.9	3.1	2.5	2.8	2.8	3.5	3.5	4.0	3.8	3.5	4.1	3.6	3.6
	PrecMax	51.9	4.7	5.0	6.0	6.4	7.4	6.4	6.0	6.6	7.0	8.0	6.1	7.4
	PrecMin	30.0	1.4	0.1	0.9	0.8	0.9	1.3	1.9	1.0	1.1	0.6	0.7	1.3
	PrecStd	5.5	1.1	1.1	1.2	1.3	1.6	1.3	1.3	1.4	1.6	2.0	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.7	41.4	47.6	61.3	74.5	81.2	84.2	83.4	78.7	66.8	56.1	44.0
		MCWB	41.1	39.2	40.4	48.6	58.7	67.2	71.1	70.4	67.6	58.5	54.3	43.1
	2%	DB	36.5	36.6	42.0	52.0	67.5	76.1	79.3	78.0	72.2	61.0	51.2	40.3
		MCWB	35.3	34.6	36.7	43.0	54.9	64.0	68.5	67.6	63.4	55.0	49.1	39.0
	5%	DB	33.1	33.5	38.7	47.9	62.3	71.7	76.2	74.7	68.4	57.2	47.6	37.2
		MCWB	31.8	31.5	35.1	41.4	52.8	61.7	66.4	65.4	62.1	53.1	45.0	35.5
	10%	DB	29.8	30.6	36.2	44.6	57.9	68.0	73.5	71.9	65.5	54.2	44.3	34.8
		MCWB	28.4	28.8	33.1	38.9	50.5	59.5	64.9	64.0	60.1	50.5	41.7	33.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	40.0	42.4	50.7	62.3	70.0	72.9	72.0	69.3	60.4	54.6	43.6
		MCDB	41.8	40.8	44.7	57.3	69.4	78.7	80.6	79.3	75.4	64.5	55.6	43.9
	2%	WB	35.6	35.1	38.2	45.5	57.4	66.4	70.5	69.3	65.9	56.7	49.9	39.5
		MCDB	36.3	36.1	40.4	49.7	63.6	72.9	76.9	75.3	69.6	59.8	51.0	40.0
	5%	WB	32.0	31.9	35.6	42.3	54.3	63.7	68.4	67.4	63.3	53.9	45.6	35.9
		MCDB	32.9	33.2	38.1	46.3	60.6	69.2	74.0	72.5	67.4	56.6	47.0	36.9
	10%	WB	28.5	28.8	33.4	39.7	51.2	61.0	66.6	65.7	60.8	51.1	42.2	33.5
		MCDB	29.8	30.5	36.0	43.7	57.0	66.4	71.7	70.0	64.5	53.6	43.7	34.5
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.2	13.9	13.2	15.9	16.7	15.9	15.8	15.8	12.6	10.5	11.1
		MCDBR	16.3	14.4	15.6	19.4	24.3	22.8	20.0	19.6	19.3	16.7	13.7	10.8
		MCWBR	16.5	12.8	12.0	12.7	14.4	13.6	11.0	10.7	12.2	12.3	12.5	10.9
	5% WB	MCDBR	16.5	13.3	14.0	16.5	21.3	19.8	17.3	17.2	17.1	15.1	12.9	11.1
		MCWBR	17.0	12.6	11.9	12.5	14.1	13.0	10.6	10.3	12.3	12.5	13.1	11.5
Clear-Sky Solar Irradiance	taub	0.275	0.279	0.313	0.382	0.368	0.386	0.422	0.394	0.361	0.334	0.299	0.279	
	taud	2.362	2.301	2.301	2.240	2.380	2.354	2.303	2.379	2.453	2.508	2.513	2.427	
	Ebn at Noon	256	281	287	275	284	278	266	269	268	259	247	240	
	Edh at Noon	22	29	34	41	36	38	39	35	30	24	19	18	
All-Sky Solar Radiation	RadAvg	400	661	1053	1376	1629	1759	1725	1514	1166	723	390	289	
	RadStd	41	69	83	115	140	137	138	94	97	57	48	30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	+3.10	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)		N/A	N/A	+2.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page